HANGCHA

user manual



Lithium battery charger

Contents

Chapter 1 Overview.....2

1.1 Instructions for use.....2
1.2 Safety warning.....2
1.3 Terms of use.....2
1.4 For the crowd.....2

Chapter 2 Safety precautions.....2

2.1 Safety Statement.....2
2.2 Safety tips.....2
2.3 Safety regulations.....3

Chapter 3 Product Introduction.....4

3.1 Referenced standards.....4
3.2 Conditions of use.....4
3.3 Product overview.....4
3.4 Intelligent charger technical configuration parameters.....5
3.5 The structure and size of the charger.....5

Chapter 4 Installation and commissioning.....5

4.1 Installation precautions.....5
4.2 Installation precautions.....6
4.3 Enter the wiring definition.....6
4.4 Output wiring definition.....7
4.5 Installation and commissioning.....7

Chapter 5 Charging operation steps.....7

Chapter 6 Check the maintenance instructions regularly.....9

Chapter 7 Coding and information.....9

Chapter 8 Common faults and troubleshooting methods.....10

Chapter 9 Warranty matters.....11

Chapter 10 Packaging, shipping, and storage.....11

Chapter 1 Overview

1.1 Instructions for use

The safety precautions, product features, technical performance, installation and debugging, fault handling, operation steps, equipment maintenance, and other contents of the lithium battery charger are expounded and explained. Please read the manual carefully after opening the box, and keep it properly for later reference.
The Company reserves the right to modify the specifications, and
has the right not to give further notice.

1.2 Safety Warning



1.3 Terms of use

★When using and connecting the lithium battery charger, the relevant regulations for electric forklift truck charging should be strictly observed;
★ The lithium battery charger has been developed, produced, inspected, and filed according to the relevant safety standards. If all prescribed instructions and safety technical instructions are followed, no property damage or harm to personal safety is normally caused;

1.4 For the crowd

This manual is intended for the following groups of people:
★ Charging operator;
★ Charging equipment training personnel;
★ Technical service personnel.

Chapter 2 Safety precautions

2.1 Safety Statement

Failure to comply with the safety precautions in this section on the use of the equipment may lead to life danger, injury, and equipment damage. Hangzhou Hangcha Yunrui Technology Co., Ltd. shall not be liable for any claims arising therefrom.

2.2 Safety Tips

To ensure personal and equipment safety, the personnel operating the equipment must comply with the following principles and regulations:
The equipment is well grounded;
★ Before the equipment is powered on, please confirm that the equipment is well grounded
★ Before the equipment is energized, confirm that the input voltage, frequency, circuit breaker or fuse, and other conditions of the device meet the specifications

★ All tools should be insulated to prevent exposed metal parts from touching the metal frame, resulting in a short circuit

★ Ensure the stable operation of the equipment. The equipment shall operate in a clean and well-ventilated environment as far as possible, and the operating environment shall not contain volatile gases or flammable gases

2.3 Safety regulations

Tips and warnings can be found in different places in this manual to indicate possible hazards, using the following symbols

1.Danger

★Electrical hazard, fire hazard! Do not use damaged, worn, or dirty charging guns;

★ Do not place inflammable, explosive, or combustible materials, chemicals, combustible steam, and other dangerous goods close to the charging equipment;

★The power supply must be cut off during the installation and replacement of the equipment to prevent electric shock;

★Do not use any extension cable to connect the charging device to the electric forklift trucks;
The charging device is only connected to the electric vehicle; do not connect to other loads (power tools), etc.

★Do not cause mechanical damage to the charging cable (bending, squeezing, or rolling), and the surface of the cable should not come into contact with heat sources, dirt, or water sources;

★All cables of the equipment must be firmly wired, well insulated, and properly sized. Loose connection, damage, or insufficient size of the cable will cause circuit failure, which may cause irreversible damage to the equipment, and even cause casualties or fire;

★Falling equipment, even if the appearance is not damaged, also can not be directly installed, because part of the internal damage may cause system disorder or circuit short circuit and other faults, must be tested by professional personnel, to determine whether it can be installed;

★Never allow the equipment to be flushed with water. The equipment in water will cause a short circuit.

2.warn

★Any fault that may affect its safety performance must be immediately eliminated by a professional before it can be restarted;

★When pulling out the charging gun, please hold the charging gun handle; do not pull the cable;

★Do not let heavy objects press down on the cable, or people step on the charging cable;

★Keep the charging gun clean and dry. If there is dirt, please wipe it with a clean, dry cloth. Do not touch the charging gun core directly with your hand;

★The products are installed by the company's professional and technical personnel (or an authorized professional organization), and unauthorized installation or modification may lead to safety accidents or equipment damage. The Company shall not be liable for any loss caused thereby.

3. Take care

★If there is any abnormal situation during use, please immediately press the emergency stop button to cut off all input and output power supply;

★In case of rain and thunder, please use it carefully;

★The normal operation of the equipment depends on the reasonable setting of some parameters, which are generally set after on-site installation and debugging. If not necessary, it is not recommended for users to reset the parameters by themselves;

★Charging can only be carried out after the vehicle is turned off,and the vehicle can only be started after charging is completed.During the charging process, electric forklift trucks are prohibited from driving;

★Regularly check whether the charging gun and charging cable are damaged, and check whether the equipment shell is damaged;
★When using charging devices to charge electric forklift trucks,please carefully read the relevant prompts and user manual of the vehicle;
★The warning and operation signs inside the equipment body and cabinet play an important role in the safe operation of the equipment. It is strictly prohibited to tear or damage them. If there is any damage or blurriness, please replace them in a timely manner.

Chapter 3 Product Introduction

3.1 Referenced standard

★Q/HCC J03045-2021 Industrial vehicle non-on-board conduction lithium battery charger;
★T/CCMA 0110-2020 Industrial vehicle non-on-board conduction charger;
★NB/T 33001-2018 Electric vehicle non-on-board conduction charger technical conditions;
★NB/T 33008.1-2018 Electric vehicle charging equipment inspection test specification Part 1: non-vehicle charger;
★GB/T 27930-2015 Communication protocol between electric vehicle non-on-board conduction charger and battery management system.

3.2 Conditions of use

★The altitude shall not exceed 2000 m;
★Operating environment temperature of the equipment: -20 ℃ ~ + 50 ℃;
★The maximum relative humidity of the surrounding air shall not exceed 95% (when the temperature of the ambient air is 25 ± 5℃);
★The installation foundation should be stable, and there should be no violent vibration or impact, and the vertical inclination should not exceed 5%;
★ The operation site has no conductive or explosive dust, no gas or vapor that corrodes metal or destroys insulation;
★ Outdoor operation requires the installation of protective sheds around and above the equipment to prevent rain.

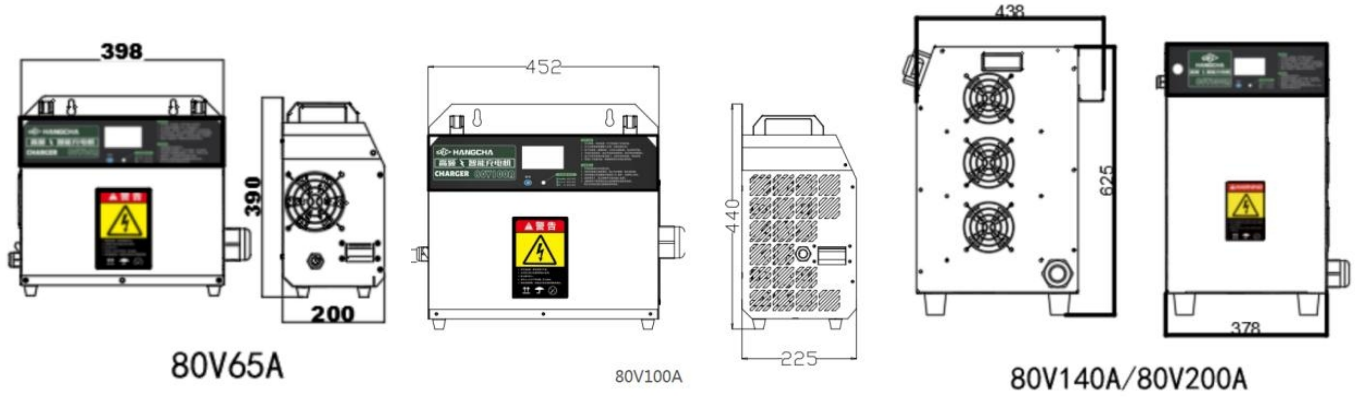
3.3 Product overview

This product has a high input power factor, low input current harmonic, high efficiency, and EMC performance, simple technology, high standardization, high reliability, beautiful appearance, simple use, operation, and maintenance, far better than the same application products in the industry, and can be widely used in transportation, industry, logistics, and other industries.

3.4 Intelligent charger technical configuration parameters

Model number	48V100A	48V200A	80V65A	80V100A	80V140A	80V200A
Rated input voltage	AC380±15% (three phase)					
Max input current	10.3A	19A	12A	16A	23.5A	31A
Enter the grid frequency	50/60Hz					
power factor	≥0.99					
Input power	≤6.2kW	≤12.5kW	≤5.2kW	≤10kW	≤15.5KW	≤19kW
Efficiency	≥0.93					
Rated output voltage	58.8V		88.8V			
Rated output current	100A	200A	65A	100A	140A	200A
Rated output power	5.7kW	11.5kW	6kW	10kW	14.2KW	18kW
DC output range	10~60Vdc		10~100Vdc			
Output connector	GB125A/Rema160A/320A					
Cooling mode	Forced air cooling					
Operating ambient temperature	-20~-50℃ full output; 50~70℃ derated output					
Relative humidity	≤95%					
Class of protection	IP20					
noise	<65dB					
above sea level	<2000m					

3.5 The structure and size of the charger



Chapter 4 Installation and commissioning

4.1 Installation precautions

The operating voltage and current of some components in the power supply system are very large. In order to ensure personal safety, the following provisions should be observed at all times:

1. Only personnel who have received power system training and have full knowledge of power systems can install this power system. The safety regulations listed above and local safety regulations should be observed at all times during installation.

2. The system voltage may reach 320Vac or 475Vac, which is a dangerous voltage. Insulation protection is required during installation and operation to ensure personal safety.
3. If you want to operate in the power system cabinet, ensure that the power system is not powered on. Disconnect the mains input and battery power from the power system.
4. Place the power supply device in a locked room, lock the cabinet, and the person in charge of the power supply device will keep the key.

4.2 Installation precautions

Model number	Max input current	Section area of the input line	Air switch Model number
48V100A	10.3A	3*4+1*2.5mm² 3P+PE	32A-3P
48V200A	19A	3*6+1*6mm² 3P+PE	63A-3P
80V65A	12A	3*2.5+1*2.5mm² 3P+PE	32A-3P
80V100A	16A	3*4+1*2.5mm² 3P+PE	32A-3P
80V140A	23.5A	3*4+1*4mm² 3P+PE	32A-3P
80V200A	31A	3*6+1*6mm² 3P+PE	63A-3P

4.3 Enter the wiring definition



Wiring definition: The input cable is divided into three-phase 4-wire specifications, and the three-phase 4-wire interfaces are L1, L2, L3, and PE (grounded). Explanation: L1,L2, and L3 are the three live wires, and PE is the ground wire. L1 shows the black line, L2 is the brown line, and L3 is the blue line. The PE is the yellow and green line, and the grounding line must be grounded.



Wiring definition: Yida interface is three-phase 4 lines, R1 S2 T3 = L1 L2 L3 = A B C = UVW (four kinds of identification are national standard identification) connected to three fire lines, PE or ground wire identification grounding wire. Installation method: the screw hole position is on the side, plug the wire into the hole, and tighten the screw from the side. 16A can connect 1-4 square lines, 32A can connect 2.5-6 square lines, 63A can connect 4-10 square lines.



Wiring definition: the interface is three-phase 4 lines, the plug front line order is: left one phase, lower two phases, right three phases, the top is the ground wire.
Warning: the PE protection land must be connected, otherwise it may threaten the safety of people and equipment. The company is not responsible for the accident caused by the user's wrong connection to the communication input line. Thank you for your cooperation.

4.4 Output wiring definition

The output line is generally a REMA connector, which is subject to the actual product.



Wiring definition: The charging gun is designed according to GB/T 20234.3-2015, and the wiring is defined as: DC power line: DC+, DC-, equipment ground line: PE, charging communication line: S+, S-, charging connection confirmation line: CC1, CC2, low voltage auxiliary power line: A+, A-.



Wiring definition: match the corresponding definition according to the customer's corresponding battery definition

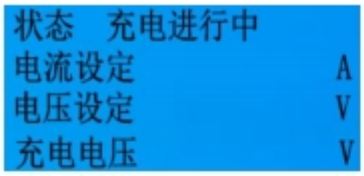
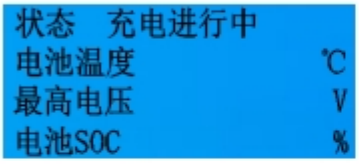
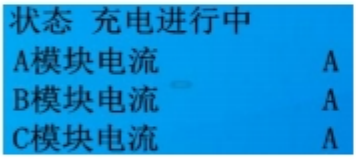
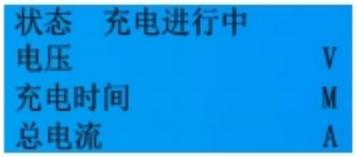
4.5 Installation and commissioning

Some components inside the charger have high working voltage and current. To ensure personal safety, the following regulations should be followed:
★Only trained and qualified personnel can install this charger;
★Insulation protection shall be provided during installation and operation;
★ Grounding measures for the charger should be taken.
★Fan air intake and fan air outlet should be at least 10 cm apart
★The charger should avoid rain
★Do not bring flammable, explosive, or combustible materials close to the recharger
★Use it with caution in thunderstorm weather

Chapter 5 Check the maintenance instructions regularly

The input and output power wiring of the charger can only be operated by qualified professionals. Connect after the input socket is unplugged or the circuit breaker is disconnected.
1) Move the forklift near the charger and make sure the plugs and hands are dry before operation.
2) Check the charger before starting charging to ensure that the charger is wired correctly without any fault or abnormal phenomenon.
3)Confirm whether the voltage and capacity of the battery to be charged match with the charger. (Please refer to the charger specifications and the battery specifications to be charged).

4) Before charging, the output plug of the charger is connected to the socket of the forklift battery, and then the charger is plugged into the power socket.
5) In the case of no fault, the charger receives the charging command (including charging voltage, charging current, and startup command) issued by the BMS of the forklift battery, and automatically starts charging the battery, and the charging indicator light flashes.
6)If an error occurs during charging, press "Pause" to stop charging until the fault is resolved.
7) After the battery is fully charged, the charger automatically stops charging, and the charger indicator light is on.
8) If you need to use a forklift when charging, press "pause" to stop charging and disconnect the plug.
9) During charging, the charging indicator blinks and the following four screens appear in sequence:

	**Current setting: Set the current sent by the battery BMS to the charger** **Voltage setting: The voltage set value that the battery BMS sends to the charger** **Charge voltage: indicates the current battery voltage**
	**Battery temperature: the temperature value during the battery charge** **Maximum voltage: The highest voltage value of a battery pack cell** **Battery SOC: Current battery charge value**
	**A Module current: the current output current value of the charger module A** **B Module current: the current output current value of the charger module B** **C Module current: the current output current value of the charger module C**
	**Voltage: the current voltage value of the battery** **Charging time: the cumulative time value during charging** **Total current: the maximum current output of the charger in operation**
	**Pause button: Press once to pause charging, then press again to resume charging** **Indicator light: full and bright, flashing while charging, standby is not bright**

10) After the charging is completed, the indicator light remains on and the display screen shows that the charging is complete. First, disconnect the AC circuit breaker of the charger, unplug the AC input plug, and finally unplug the battery connection plug.
Note: Non-standard operation will cause damage to the battery or charger, or other unexpected risks

Chapter 6 Check the maintenance instructions regularly

★Whether the input and output wires are damaged (test before charging);
★Whether the output plug of the charger is damaged (test before charging);
★If the above situation occurs, please replace the relevant accessories in time (test before charging);
★Clean up the dust and other foreign bodies on the charger regularly (check once a month);
★Confirm that the charger and the charger input circuit breaker are working normally. If any bad situation is found, please contact the sales store or the factory in time;
★Cut off the input power supply before the bad cause is clear, and then unplug the charging input AC plug to avoid electric shock

Chapter 7 Coding and information

encoding	Encode the corresponding information	encoding	Encode the corresponding information
F4	Insulation fault	F14	Stop for untrustworthy reasons
F5	Output connector overtemperature	F15	The monomer voltage is too high
F6	Vehicle BMS overtemperature	F16	The monomer voltage is too low
F7	Charging connection failure	F17	Charging overcurrent
F8	Battery pack temperature is too high	F18	Battery overtemperature
F9	High-voltage relay failure	F19	Insulation anomaly
F10	Detection point 2 detects the fault	F20	Connector failure
F11	Other faults		
F12	Current anomaly		
F13	Voltage anomaly		

Chapter 8 Common faults and troubleshooting methods

number	Fault phenomenon	Failure cause	resolvent
1	Fan fault	Fan is stuck, damaged, or the fan The plug-in is loose.	Check the fan net cover for deformation or damage, Whether the fan connector is loose and internal Accumulate the amounts of dust or foreign matter.
2	Lack of phase	Enter one phase of three phases.	Check whether AC input voltage is normal and transmission Whether the entry plug is loose.
3	Input undervoltage	Input voltage is below 310V.	Check whether the AC input voltage is normal and the input plug is loose.
4	communication failure	Abnormal CAN signal line or a BMS Data not sent.	Check that the communication cable between the BMS and the charger is properly connected.

Chapter 9 Warranty Matters

9.1 The product is guaranteed for one year (the company does not bear the logistics, transportation, customs duties, and other costs caused by maintenance). If the warranty period expires, the user shall negotiate with the manufacturer to take a paid way for replacement or maintenance.
9.2 In the process of using our products, if there is any problem, please check according to this manual first. If still unable to resolve, please contact our company
9.3 In the process of using the company's products, if the following situations occur, the company does not assume the warranty obligation.

★ Product damage caused by use in violation of product manuals or specifications
★ Damage to the products caused by unauthorized modification or self-repair
★ Damage to the product due to an abnormal external force.
★ Damage to products due to natural disasters or other force majeure

9.4 For the quality problems caused by the scope of the warranty, the company's responsibility is limited to the product itself, and it does not assume joint and extended liability
9.5 Matters needing attention when abandoned
The waste disposal of the charger is industrial waste. Please hand it over to certified professionals or companies for disposal. The company is not responsible for recycling.

Chapter 10 Packaging, shipping, and storage

★ The charger is packed with carton and anti-collision material.
★ Products during transportation should not be subjected to violent vibration, impact, inversion, etc
★ The charger storage should avoid high temperature and humidity, dust, metal powder, and other environments to avoid the coexistence of acidic, alkaline, or other corrosive gases. Store in a dry, well-ventilated indoor environment
★ There should be pallets and other moisture-proof measures on the storage floor.

HANGCHA

This manual's content may be subject to change without notice, and the final interpretation rights belong to Hangzhou Hangcha Yunrui Technology Co., Ltd.

HANGCHA GROUP CO.,LTD

Company address:No. 666, Xiangfu Road, Qingshanhu Street, Lin'an District, Hangzhou